

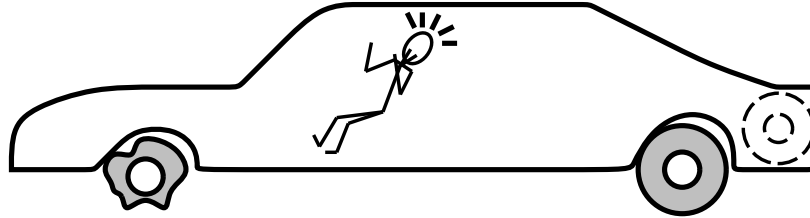
Planning and Acting

CHAPTER 13

Outline

- ◇ The real world
- ◇ Conditional planning
- ◇ Monitoring and replanning

The real world



START

*~Flat(Spare) Intact(Spare) Off(Spare)
On(Tire1) Flat(Tire1)*

On(x) ~Flat(x)

FINISH

On(x)

Remove(x)

Off(x) ClearHub

Off(x) ClearHub

Puton(x)

On(x) ~ClearHub

Intact(x) Flat(x)

Inflate(x)

~Flat(x)

Things go wrong

Incomplete information

Unknown preconditions, e.g., *Intact(Spare)*?

Disjunctive effects, e.g., *Inflate(x)* causes

$\textit{Inflated}(x) \vee \textit{SlowHiss}(x) \vee \textit{Burst}(x) \vee \textit{BrokenPump} \vee \dots$

Incorrect information

Current state incorrect, e.g., spare NOT intact

Missing/incorrect postconditions in operators

Qualification problem:

can never finish listing all the required preconditions and possible conditional outcomes of actions

Solutions

Conditional planning

- Plan to obtain information (**observation actions**)

- Subplan for each contingency, e.g.,

- $[Check(Tire1), \mathbf{If}(Intact(Tire1), [Inflate(Tire1)], [CallAAA])]$

Expensive because it plans for many unlikely cases

Monitoring/Replanning

- Assume normal states, outcomes

- Check progress *during execution*, replan if necessary

Unanticipated outcomes may lead to failure (e.g., no AAA card)

In general, some monitoring is unavoidable

Conditional planning

$[\dots, \mathbf{If}(p, [then\ plan], [else\ plan]), \dots]$

Execution: check p against current KB, execute “then” or “else”

Conditional planning: just like POP except

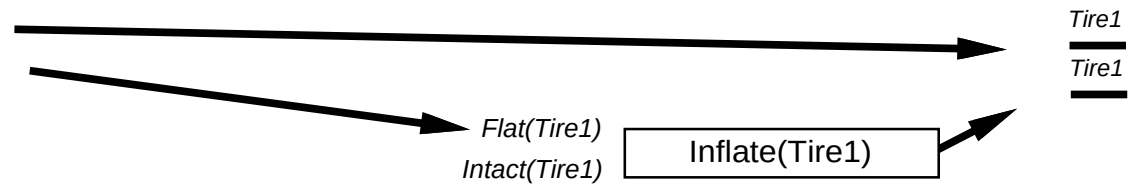
- if an open condition can be established by observation action
- add the action to the plan
- complete plan for each possible observation outcome
- insert conditional step with these subplans

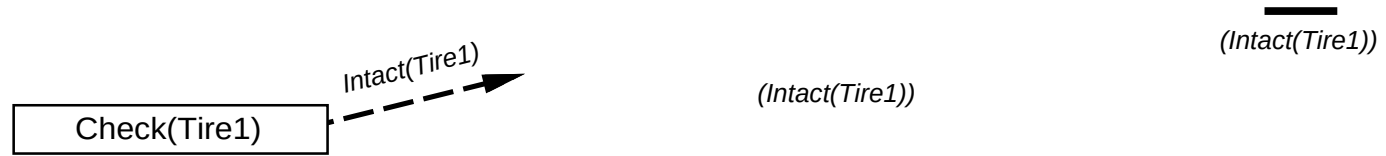
CheckTire(x)

KnowsIf(Intact(x))

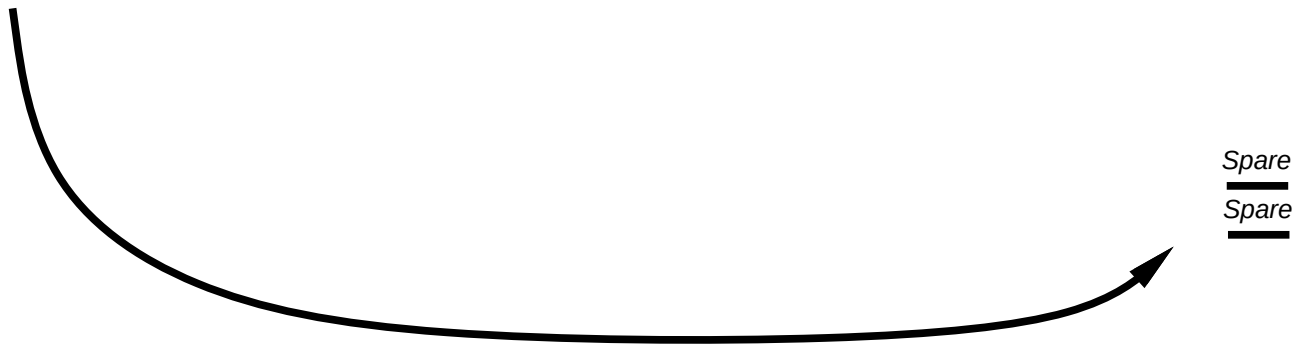
Conditional planning example

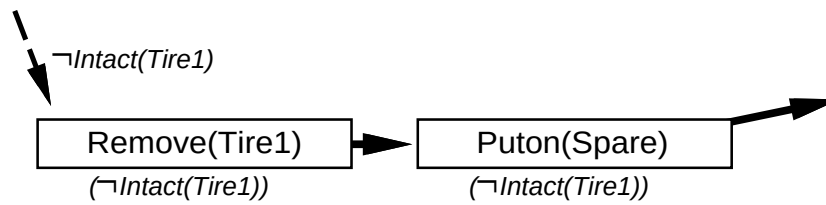






$On(x)$
 $Inflated(x)$ **Finish**
 $(\neg Intact(Tire1))$





Monitoring

Execution monitoring

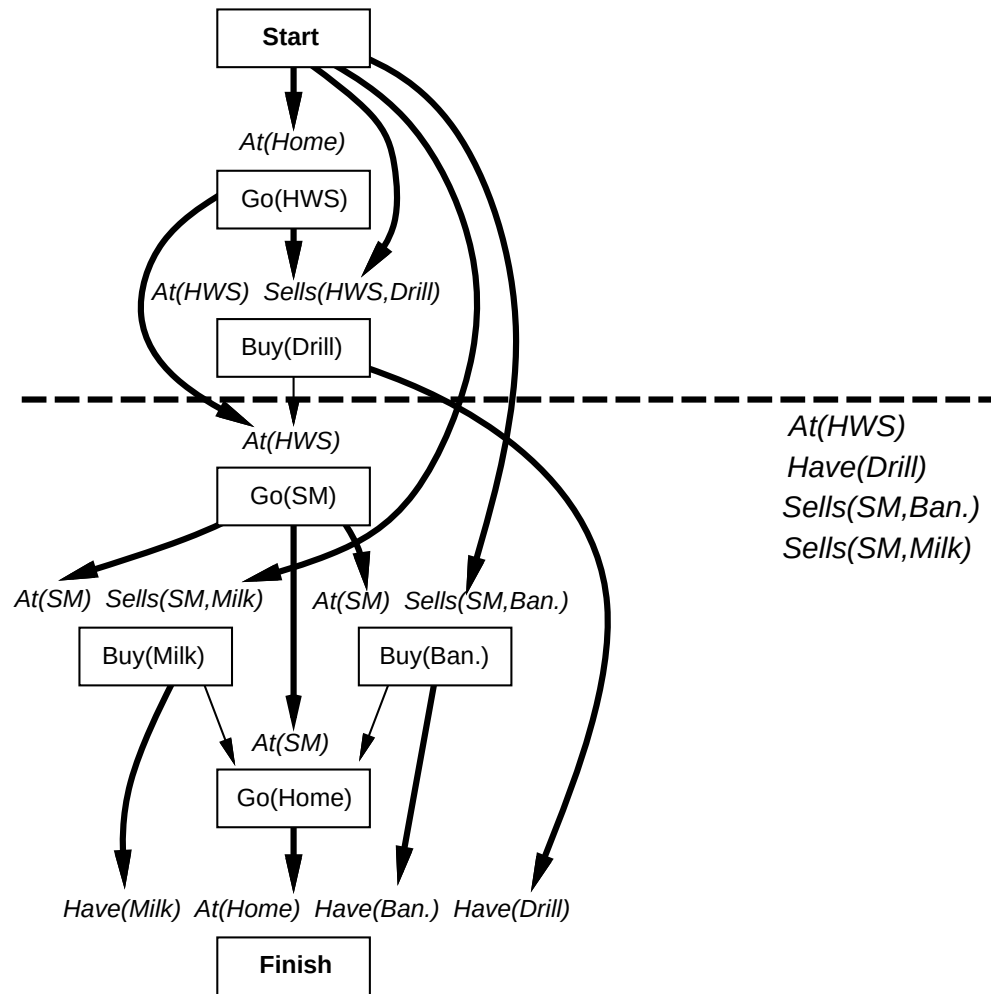
“failure” = preconditions of *remaining plan* not met
preconditions = causal links at current time

Action monitoring

“failure” = preconditions of *next action* not met
(or action itself fails, e.g., robot bump sensor)

In both cases, need to *replan*

Preconditions for remaining plan



Replanning

Simplest: on failure, replan from scratch

Better: plan to get back on track by reconnecting to best continuation
Generates “loop until done” behavior with no explicit loop

